

Heat Sink Design for WBG Power Modules Based on Fourier Series and Evolutionary Multi-Objective Multi-Physics Optimization

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This work was supported by the U.S. Department of Energy's (DOE's) Vehicle Technologies Office Electric Drive Technologies Program. The authors thank Ms.

Susan Rogers of DOE for her support and guidance, Jon Wilkins and Randy Wiles at Oak Ridge National Laboratory for CAD design and developing the experimental setup. Part of this publication was presented at 2020 IEEE Energy Conversion Congress and Exposition (ECCE) in Detroit, Michigan, USA, with the title, Fourier Analysis-Based Evolutionary Multi-Objective Multiphysics Optimization of Liquid-Cooled Heat Sinks. doi:10.1109/ECCE44975.2020.9235943 This manuscript has been authored by UT-Battelle, LLC, under Contract DE-AC05-00OR22725 with the U.S. Department of Energy (DOE). The U.S. government retains and the publisher, by accepting the article for publication, acknowledges that the U.S. government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this manuscript, or allow others to do so, for U.S. government purposes. DOE will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan.

ABSTRACT Optimal heat dissipation in power modules can significantly increase their power density. Removing the generated heat is critical for capturing the benefits of advanced semiconductor materials and improving the reliability of the device operation. This study proposes a design optimization method for liquid-cooled heat sinks that use a Fourier analysis-based tool and an evolutionary optimization algorithm to optimize the heat sink geometry for specified objectives. The optimized heat sink geometry was compared with state-of-the-art solutions in the literature based on finite element analysis of different designs. The proposed methodology can develop complex geometries that outperform conventional heat sink geometries. Optimized heat sink design from the proposed method was fabricated and tested in an experimental setup under representative operating conditions. The experimental setup was also modeled in the finite element model that was used for the proposed heat sink optimization method. The experimental results show that developed finite element models can predict the thermal and flow performance of the complex design with high fidelity, and the results validate the proposed design approach.

INDEX TERMS Evolutionary algorithms, heat sink, multi-objective optimization, power module, thermal management.

I. INTRODUCTION

Transportation electrification necessitates improvements in system efficiency and power density. Achieving a high power density not only requires advanced power conversion modules that use the most efficient semiconductor devices, but also optimal thermal management systems [1]–[3]. Including wide bandgap-based semiconductor devices in power modules raises the power conversion efficiency [2], [3]. However, these devices still exhibit substantial power losses in a small volume. In the absence of adequate heat dissipation from the semiconductor devices, their performance degrades, and the

devices fail to continue their intended operation. To improve the power density of power modules, electrical and thermal optimization must be performed since the two subsystems closely interact with each other. Without proper thermal optimization, the benefits of the advanced power semiconductor devices and packaging schemes cannot be realized.

For the high-power density applications being considered, liquid cooling of power modules rapidly removes the generated semiconductor heat. The topology optimization of such heat sinks to achieve low-volume high-power density modules is an active area of interest. One category of

heat sink design optimization involves gradient/Hessian-based methods [4]–[6]. Another category is based on the use of population-based evolutionary algorithms [7]–[9]. Evolutionary optimization algorithms have advantages, such as their simplicity in problem setup and inclusion of various kinds of constraints (e.g., linear, piece-wise, nonlinear), their lower susceptibility for being trapped at a local extrema, and no gradient computation. Hence, this study considers evolutionary optimization algorithms for heat sink design.

In the case of structural design optimization, the number of design variables can become infinitely large during problem formulation [9]. Because evolutionary algorithms rely on population quality for time-efficient optimization, inferior population could lead to the computationally inefficient execution of the optimization or sub-optimal solution for the considered problem. Improper problem formulation leads to difficulties in the convergence of optimization engines [10].

As discussed above, population generation is critical for evolutionary algorithms to provide high quality results with reasonable computational resources. Large number of design variables, poor population quality and inferior populations should be avoided to achieve high quality designs with reasonable computational resources and time. To address the issues in population generation and optimization, a Fourier analysis-based geometry representation tool is proposed. The proposed tool can capture the details of heat sink geometries and can generate distinct heat sink profiles with few variables (i.e., Fourier parameters). The proposed tool was used in a population-based optimization engine to optimize the heat sink geometry for specified objectives using finite element analysis (FEA).

This article is organized as follows. Section II discusses issues in heat sink representation for the evolutionary optimization algorithm and proposes a Fourier analysis-based heat sink geometry representation tool. Next, using the geometry representation, the heat sink design problem is formulated in Section III, which discusses assumptions in the problem, design search domain, constraints to be imposed, and optimization objectives. The genetic algorithm (GA) was used as the optimization engine; a brief overview of its procedure is presented in Section IV. Using the formulated problem and optimization engine, a case study was undertaken as described in Section V, and the results from the optimization are compared with literature. The experimental validation of an optimized heat sink is discussed in Section VI. Finally, conclusions and a future work are discussed in Section VII.

II. FOURIER ANALYSIS-BASED HEAT SINK GEOMETRY REPRESENTATION

To optimize heat sinks geometries, the heat sink structures should be represented using the fewest variables possible. This necessity stems from the mode of operation of evolutionary algorithms, which typically rely on a large population for their operation [10]. Reducing the number of design variables for each population helps reduce overall computational complexity. For heat sink optimization using population-based

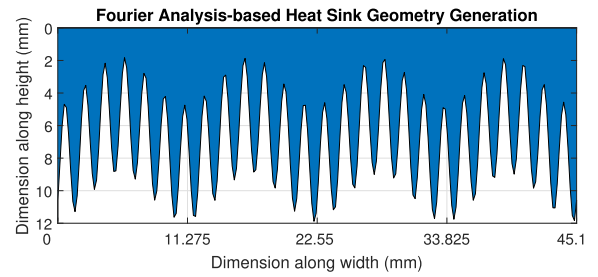


FIG. 1. Illustration of Fourier analysis-based geometry.

algorithms, binary matrix representation has been investigated in the literature [9]. In the scheme presented by Dokken *et al.* [9], a value of 0 in the matrix represents no heat sink, and 1 represents the presence of heat sink material. In such representations, the model processing speed and accuracy depend on the number of elements used in the matrix [9] and are unsuitable for evolutionary algorithms.

Another method considered in this article is to represent the geometry by using orthogonal functional space wherein the geometry is represented as a linear combination of the orthogonal functions. A higher resolution of the structural data representation can be achieved by using fewer variables in this representation scheme, unlike the binary case. Of the available functional spaces, sinusoidal orthogonal space is of particular interest because of its simplicity in application and its widely available literature and analysis tools.

For example, consider a 2D surface H confined in an area spanned by $\{0 \leq x \leq L_x, 0 \leq y \leq L_y\}$ in a Cartesian coordinate system, where L_x and L_y are the maximum allowed horizontal and vertical dimensions, respectively. The 2D structure is created starting from its base at $y = 0$. For simplicity, the structure is assumed to have no holes and can be mathematically defined using a single dimensional function with each point representing the height of the material added to the structure. By using the mathematical representation of a stationary wave, the surface of the manufactured structure $F_{am}(x)$ —considering the assumptions—can be expressed using the summation of sinusoidal harmonics as

$$F_{am}(x) = F_0 + \sum_{n=1}^{N_{hs}} \left(F_h[n] \cos \left(\frac{2\pi}{\lambda_x} h[n]x + \phi_h[n] \right) \right), \quad (1)$$

where $x \in [0, L_x]$; F_0 is a constant shift; λ_x is the wavelength, which is also equal to L_x ; $h[n]$ is the harmonic order; $F_h[n]$ and $\phi_h[n]$ are its amplitude and phase shift, respectively; and N_{hs} is the total number of harmonics considered.

The total number of variables in (1) is equal to $(3N_{hs} + 1)$, including the shift F_0 in the structure. An illustration of the harmonic geometry generation is shown in Fig. 1, wherein $N_{hs} = 2$, $L_x = 45.1$ mm, $F_0 = 6.85$ mm, $h = [4 \ 26]$, $F_h = [0.74 \ 1.8]$ mm, and $\phi_h = [353 \ 309]$ degrees. The structure is shown upside down in the figure to resemble a typical heat sink. By using just two harmonics, a fairly complex heat sink structure is represented.

Finally, to create a 3D structure, the 2D surface is extended along the third dimension, assuming the same cross-sectional

structure of the constructed 2D surface. Using this geometrical representation, a heat sink design optimization method was developed. An optimization engine will optimize the Fourier parameters to develop structures for the best possible performance metrics.

III. FORMULATION OF THE HEAT SINK DESIGN OPTIMIZATION PROBLEM

To formulate the problem, the set of variables must first be defined. In this problem formulation, it is assumed that (1) the power module semiconductor device layout is fixed, (2) heat sink material parameters are known, and (3) coolant material and flow conditions are fixed. With known conditions, the structural design of the heat sink remains as the unknown for the considered optimization. By using the Fourier analysis-based geometric representation described in Section II, the harmonic content in the heat sink structure creates the variable vector with a size of $(3N_{hs} + 1)$ parameters.

A. HEAT SINK PERFORMANCE COMPUTATION

The power module heat sink was optimized at its worst-case steady-state thermal conditions. That is, the heat loading conditions used for the heat sink optimization were taken for a power module at its peak power/current load by operating in steady-state condition.

The cooling performance of the heat sink is measured using metrics obtained from multi-physics FEA studies. Steady-state FEA simulations were performed to simulate the steady-state heat transfer and coolant flow in the heat sink. To reduce the computation time involved in flow simulation, the flow conditions were assumed to remain fully developed and laminar. Few geometrical constraints were imposed to ensure the developed structures meet requirements, and these constraints were (1) a limit on the maximum allowed heat sink height and (2) a limit on the maximum allowed Reynolds number. The first constraint limits the maximum possible heat sink volume, whereas the second constraint limits the flow to be laminar.

To further reduce the computational time complexity, the coolant properties were assumed to be temperature invariant. Furthermore, the coolant flow rate was assumed to be fixed at the inlet of the heat sink. From the assumptions, the fluid flow and heat transfer problem can be considered weakly coupled. Hence, the coupled multi-physics simulation was split into two single-physics simulations wherein the coolant flow in the heat sink was first computed by assuming a fixed temperature, and then the heat transfer simulation was performed using the solved coolant flow. A flowchart of an FEA solution scheme is shown in Fig. 2. For increased accuracy of the simulations, the flow and heat transfer simulations can be fully coupled at the expense of increased computational time. Such simulation can be useful for cases in which the temperature increase across the coolant is significant enough to change the thermo-mechanical properties (e.g., thermal conductivity, viscosity).

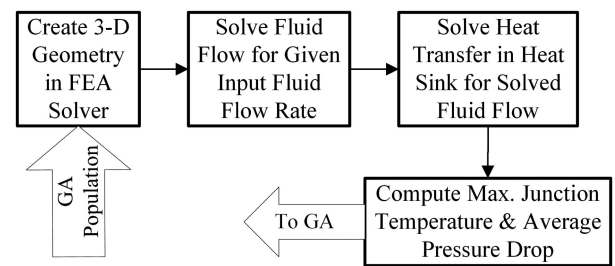


FIG. 2. FEA solution scheme for the proposed optimization approach.

B. OPTIMIZATION OBJECTIVES AND CONSTRAINTS

After the FEA, metrics of interest are evaluated to determine the design fitness for the optimization engine. In this study, minimizing the heat sink volume and coolant pressure drop was desired to achieve high power density and enable maximum use of wide-bandgap device properties in the system. The volume minimization increases the power density of the power module, and the pressure drop minimization reduces the pumping power requirement. As mentioned, the GA was used as the optimization engine [10], [11]. Furthermore, certain design constraints were imposed to limit the design search space and avoid passing on unfeasible designs to the next generation population. Such constraints can be maximum allowed junction temperature of the power semiconductor die, dimensional limits for manufacturability of the design, and Reynolds number.

C. OPTIMIZATION PROCEDURE

The design optimization began with a GA population generation containing Fourier parameters, as shown in (1). Using this harmonic population and (1), a 2D cross-section of a heat sink was created. The GA then passed this information to the structure formation tool, which extended the 2D cross-section in length by assuming a fixed shape for the creation of the 3D heat sink structure. The heat sink structure was then combined with the power module structure and passed to the FEA tool for computation of its steady-state heat transfer and coolant flow performance. After FEA, the heat sink was verified against imposed constraints. If successful in passing the constraints, the GA takes over the optimization process and uses the simulation results to compute heat sink performance metrics using factors such as coolant pressure drop, device temperatures, and volume. The performance data are used to determine the design fitness, which GA operations then use to create an optimal heat sink design population. At the end of the optimization, the solution of the multi-objective design optimization is a Pareto-optimal front of designs. A flowchart for the overall optimization process is shown in Fig. 3.

IV. GA AND MULTI-OBJECTIVE OPTIMIZATION

This section presents a brief overview of the GA and its procedure used in this study. A majority of this overview is based upon Sudhoff [10] and the Genetic Optimization System Engineering Toolbox [11], and these works provide

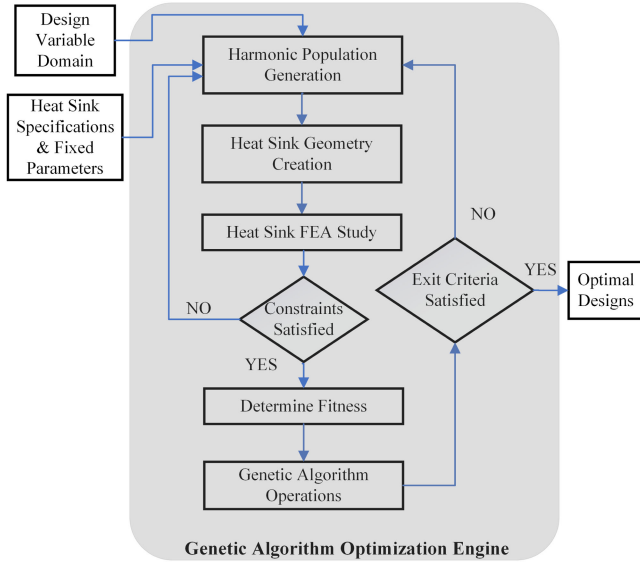


FIG. 3. Heat sink multi-objective design optimization.

further details. As is typical for population-based evolutionary optimization algorithms, GA relies on its past and current population to determine the best course of action for the future population using its operators. Consider a GA optimization problem that runs for n generations with each generation operating on m solutions (population size). In the heat sink design optimization problem, Fourier parameters are the design variables, which are stored in vector $\mathbf{v}_{g,j}$ to represent a single population element j , as

$$\mathbf{v}_{g,j} = [F_0, h[1], h[2] \dots, h[N_{hs}], F_h[1], F_h[2] \dots, F_h[N_{hs}], \phi_h[1], \phi_h[2] \dots, \phi_h[N_{hs}]]^T, \quad (2)$$

where $j \in \{1, 2, \dots, m\}$. This design vector $\mathbf{v}_{g,j}$ can also be viewed as the genetic sequence (i.e., DNA) of the population element.

The GA population at its generation α , $\mathbf{P}_{g,\alpha}$, is stored by using

$$\mathbf{P}_{g,\alpha} = [\mathbf{v}_{g,1} \quad \mathbf{v}_{g,2} \quad \dots \quad \mathbf{v}_{g,m}], \quad (3)$$

where $\alpha \in \{1, 2, \dots, n\}$, and $\mathbf{v}_{g,j}$ is the j^{th} population element. Matrix $\mathbf{P}_{g,\alpha}$ has size $[(3N_{hs}+1), m]$, where the number of rows $(3N_{hs}+1)$ is equal to number of design variables for each heat sink, and the number of columns m is equal to the population size or the number of candidate heat sinks in each population.

For a given solution search space, the GA procedure starts by creating its initial/first population $\mathbf{P}_{g,1}$, which is generated randomly to capture different regions in the search space. This initial population is then passed to the computational analysis stage to compute its performance. The computational analysis stage first uses the aforementioned structure formation tool to create the heat sink for each member of the population and then evaluates its performance using FEA. At the end of

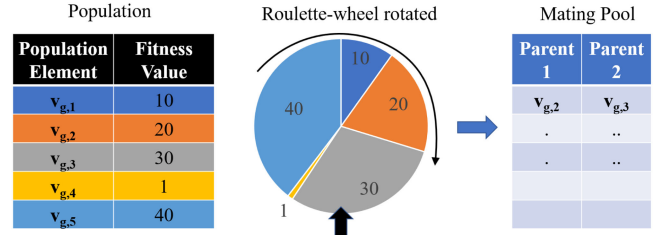


FIG. 4. Roulette-wheel selection method.

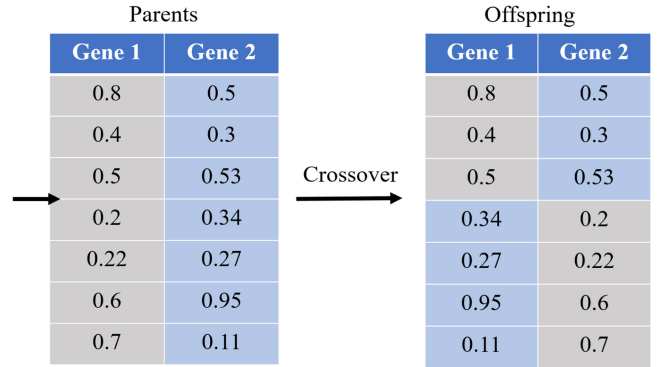


FIG. 5. Single-point crossover operation between two genes.

FEA, the computational analysis stage supplies the results to the GA.

Next, with the computational analysis results, the GA uses the “selection” operation to create a mating pool by using population elements with better fitness values. A roulette-wheel selection method is used for this operation. The purpose is to create a mating pool by using the existing population with selection probability proportional to their fitness value. High-fitness population elements have a higher chance of being selected in the pool. The process is repeated until the mating pool is full. An example of the selection process that uses the roulette-wheel method is shown in Fig. 4.

Using the mating pool, the “crossover” operation was performed in which two parent population elements create two offspring using genetic crossover. That is, the genetic sequence of one parent is crossed with the genetic sequence of the other parent by using a single- or multiple-point crossover to yield two new population elements. Consider gene $\mathbf{v}_{g,1}$ and $\mathbf{v}_{g,2}$ using (2). In a single-point crossover, a random location identifier between 1 and $3N_{hs} + 1$ is selected, and the contents of the two gene sequences are swapped to create two new offspring. An example for the single-point crossover operation is shown in Fig. 5, wherein the genetic information is crossed at location 4. Multipoint crossover works in a similar way by selecting two or more random locations in a gene and crossing over the genetic information.

Next, to mimic the genetic evolution, a “mutation” operation was performed. With probability p_m , a gene was mutated so that its genetic sequence $\mathbf{v}_{g,j}$ is modified at a single or multiple positions. At this stage, the “elitism” operator was

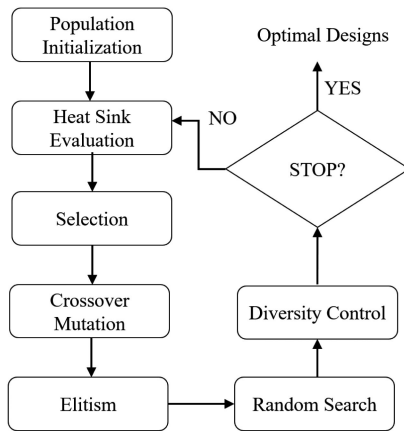


FIG. 6. GA operations.

also used to preserve the best genes for the next population, making sure that they are not altered significantly. A few additional GA operators were used, such as “diversity control” and “random search” to control and preserve the population diversity. Additional details on the aforementioned operators are available from Sudhoff and Purdue University [10], [11]. At the end of GA operations, a new population was created that went through a verification stage to determine whether GA optimization should continue. Verification was based on the generation count so that the optimization process repeated until n generations. If the generation number was less than n , then the heat sink evaluation and the same set of GA operators were performed again to create another new population. A flowchart of the GA operations is shown in Fig. 6.

In the case of a multi-objective optimization problem wherein the objectives of optimization oppose each other, multiple non-dominating solutions exist, termed as the *Pareto-optimal front of designs*. For the considered heat sink design problem, reducing the heat sink volume decreases the available coolant volume. For a given module length, the coolant volume reduction also reduces the coolant cross-sectional area, leading to an increased coolant pressure drop across the heat sink. Thus, the objectives to minimize heat sink volume (Objective 1) and minimize pressure drop (Objective 2) counteract each other. The goal of the multi-objective optimization is to find solutions the non-dominated solution set in the feasible solution space. For example, consider the solution region and solutions in Fig. 7, which shows how well Objectives 1 and 2 are achieved. The samples on the Objectives 1 and 2 plane demonstrates the available solution region, and the blue points in the region are the feasible solutions. Because the goal is to minimize both objectives, the solutions in set $S = \{a', b', c', d'\}$ (blue in Fig. 7) outperform all other designs.

However, of the four designs in set S , none is better than the other. Design a outperforms all other designs in terms of objective 1, but it does not do well in terms of objective 2 when compared with designs b , c , and d . Similarly, design b performs better than designs c and d , but it does not perform

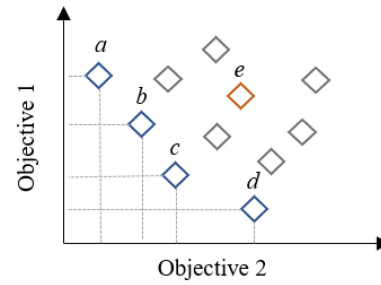


FIG. 7. Multi-objective optimization and its solutions.

better than design a in terms of objective 1. Design b performs better than design a in terms of objective 2. Through similar arguments, it is clear that none of the designs in S outweighs another in all the objectives.

Next, consider design e to be included in S . With respect to design b , e is worse in both objectives; thus, design b dominates e . Set S is a non-dominated set of solutions in which no design is better than the other; thus, design e cannot be included in S . Similar reasoning can be applied for rest of the nonoptimal designs. By using this non-dominated selection scheme, the Pareto-optimal front of the designs for a multi-objective optimization problem can be found.

A. OPTIMIZATION ENGINE AND DESIGN FITNESS

In contrast to a derivative-based optimizer, GAs do not require derivative/hessian computations and are generally immune to converging at nonglobal local extrema [10]. In this work, the GA-based tool GOSET is used as the optimization engine [11].

In accordance with the selected optimization engine, the design fitness function f is defined as [10], [11]

$$f = \begin{cases} \epsilon \begin{bmatrix} 1 & 1 \end{bmatrix}^T \left(\frac{C_s - N_c}{N_c} \right) & C_s < C_I \\ \begin{bmatrix} \frac{1}{P_{drop}} & \frac{1}{V_{pm}} \end{bmatrix}^T & C_s = C_I \end{cases}, \quad (4)$$

where

$$C_s = \sum_{i=1}^{N_c} c_i, \quad (5)$$

and where P_{drop} and V_{pm} are average pressure drop across heat sink and volume of power module respectively. N_c , C_s , and C_I are the total number of constraints, number of constraints satisfied, and number of constraints imposed during the evaluation of the objective function, respectively; c_i is the i^{th} constraint; and ϵ is a small positive number of the order 10^{-6} .

Computation of c_i is done such that if it is a less-than-equal-to constraint of form $x \leq x_{mx}$ [10], [11],

$$c_i(x, x_{mx}) = \begin{cases} 1 & x \leq x_{mx} \\ \frac{1}{1 + x - x_{mx}} & x > x_{mx} \end{cases} \quad (6)$$

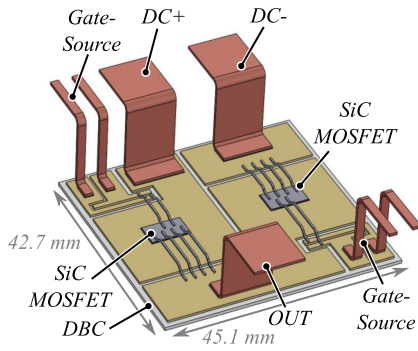


FIG. 8. DBC-based SiC half-bridge module.

otherwise, for a greater-than-equal-to constraint of form $x \geq x_{mn}$ [10], [11],

$$c_i(x, x_{mn}) = \begin{cases} 1 & x \geq x_{mn} \\ \frac{1}{1 + x_{mn} - x} & x < x_{mn} \end{cases} \quad (7)$$

If constraint i is satisfied, $c_i = 1$; otherwise, $c_i < 1$. If all the constraints are not satisfied, the objective function will yield a small negative number from (4) and (5). Otherwise, the inverse of the design metrics will be calculated in (4). These forms of constraint and fitness functions are advantageous for optimization-based designs [10]. Using GOSET, the fitness function defined in (4) is maximized [11].

V. CASE STUDY AND RESULTS

A. POWER MODULE, SIMULATION, AND OPTIMIZATION PARAMETERS

A design optimization case study was undertaken with a direct-bonded copper (DBC)-based wide-bandgap semiconductor module layout shown in Fig. 8. The design contains one 900V, 10 m Ω (CPM3-0900-0010A) SiC MOSFET die per switch location, each generating 100W power loss. The dies were attached on the DBC with 50 μm thick 63-Sn/37-Pb-based solder. To contain the coolant, 0.5mm thick plates were assumed to be attached to the two sides and the bottom side of the heat sink. Aluminum was selected as the heat sink material, which is commonly used for power electronics thermal management [7], [12]. Water ethylene-glycol mix with 50%-50% volume mixture was used as the input coolant and was supplied at (10/6) L/min flow rate at 65 $^{\circ}\text{C}$. The flow rate value is based on the assumption that six power modules will share 10 L/min flow rate for a traction drive inverter [2], [13]. In the design domain, $N_{hs} = 10$ and each harmonic order $h[n]$ is a variable between 1 and 100. The maximum allowed heat sink height is 12.5mm. Hence, minimum and maximum amplitudes of each harmonic are set as 0 and 12mm, respectively. The phase angle for each harmonic is allowed to vary between 0 and 2π . Constraints are imposed so that the total height of the heat sink is greater than 1mm and less than 12.5mm, and the Reynolds number is $\leq 2,000$.

It should be noted that although a DBC-based SiC half-bridge module was used as the case study in this design,

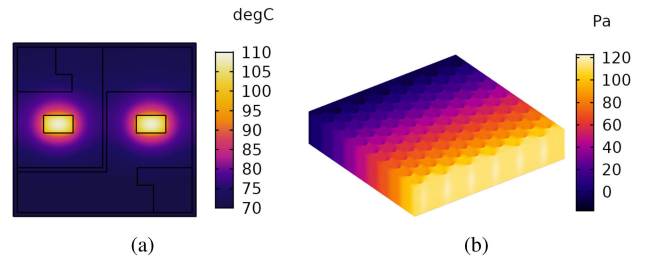


FIG. 9. (a) Steady-state thermal performance (maximum junction temperature is 108.8 $^{\circ}\text{C}$) and (b) pressure drop of the pin-fin heat sink.

proposed heat sink design approach can easily be employed with commercial power modules and Si IGBT based systems. The designer can use the CAD model of the power module they intend to use in the power converter; provide material properties, operating conditions and system constraints to the optimization engine. Based on the user input and defined optimization targets, the optimization engine will search for the design space and find the pareto-front of heat sink designs.

B. BASELINE PIN-FIN HEAT SINK FOR COMPARISON

For comparison, a pin-fin-based heat sink was taken from a 2016 BMW i3 FS800R07A2E3 power module [12], [14]; one-third of the heat sink structure was simulated in earlier work, and the performance of the pin-fin based design was discussed by Sahu et al. [15]. The pin-fin heat sink was 12mm tall, with 4mm base plate thickness and 8mm pin height. Pin radius was 2.28mm and the pin spacing information can be extracted from the module datasheet in [14]. The pin-fin-based heat sink was used with the module layout shown in Fig. 8 for its performance comparison with optimized designs.

For the selected baseline pin-fin heat sink, COMSOL FEA steady-state simulations were performed to compute its performance with the power module for the heat and coolant conditions described previously. Results from the simulations are shown in Fig. 9(a) and 9(b). As shown in the figures, the maximum device steady-state temperature is 109 $^{\circ}\text{C}$ with an average pressure drop of 120Pa. The total volume of the heat sink is 23 cm^3 . With this as the baseline, an additional constraint is imposed on the design optimization so that the optimal designs have a maximum semiconductor junction temperature of less than or equal to 115 $^{\circ}\text{C}$. This constraint will ensure that the optimal designs have better thermal performance than the baseline pin-fin heat sink.

C. HEAT SINK OPTIMIZATION RESULTS

By using the formulated design representation and FEA setup, a constrained multi-objective optimization study was performed that minimized the volume and average pressure drop across heat sink. The GA used a population size of 50 and ran for 40 generations. COMSOL was used as a heat sink computational FEA solver, which solved its coolant flow and heat transfer using the scheme shown in Fig. 2. GA operations were performed using GOSET on MATLAB.

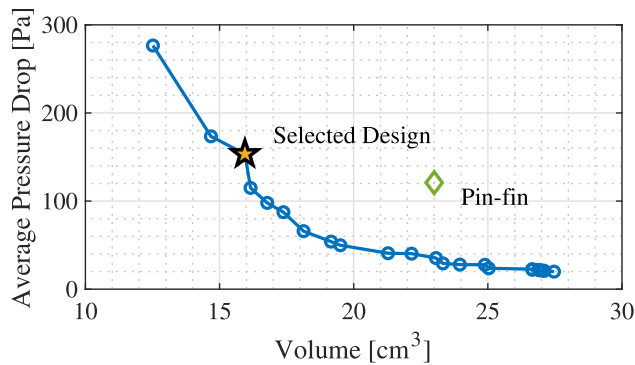


FIG. 10. Pareto-optimal front of designs.

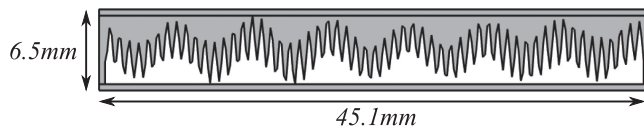


FIG. 11. Cross-section of the selected heat sink geometry presented in Fig. 10.

If a brute-force method is used to determine optimal solutions, it would take at least $(100! \cdot 10^{21} / 90!) = 6.2816 \cdot 10^{40}$ heat sink computations to be able to achieve optimal results in the absence of any optimization engines. In contrast, for this optimization, just 1,250 heat sink design candidates were evaluated by the GA-FEA setup, which significantly reduced the computational demand compared with the brute-force method. For the GA-FEA setup, the total optimization computation time on a Windows 10 system using two Intel Xeon 2.6GHz Octa-core processors was about 100h.

From the GA design optimization results, the Pareto-optimal front of the designs is shown in Fig. 10 as circled data points and is compared with the pin-fin heat sink. Optimized designs achieved volumes between 12.5 and 27.47cm³. The optimized designs achieved up to 54% volume reduction in comparison to pin-fin heat sink. The selected design from the optimization study, which will be compared against the pin-fin design, is highlighted with a star in the Pareto-optimal front. The cross-section of the selected heat sink geometry is presented in Fig. 11.

Further study of the selected design in Fig. 10 was performed. The steady-state performance of the selected heat sink is shown in Fig. 12(a) and 12(b). By comparing the thermal results of the pin-fin heat sink in Fig. 9(a) and the optimized heat sink in Fig. 12(a), the optimized heat sink clearly maintains a similar thermal performance in 30% smaller volume in comparison to pin-fin heat sink, with a maximum temperature increase of less than 3%. Figs. 9(b) and 12(b) also show that the steady-state pressure drop of the selected optimized heat sink increased 25% from 120Pa for a pin-fin heat sink to 150Pa for an optimized heat sink. The additional pressure drop of 30Pa adds a negligible extra pump power requirement for the heat sink [16]. Hence, the additional pumping power

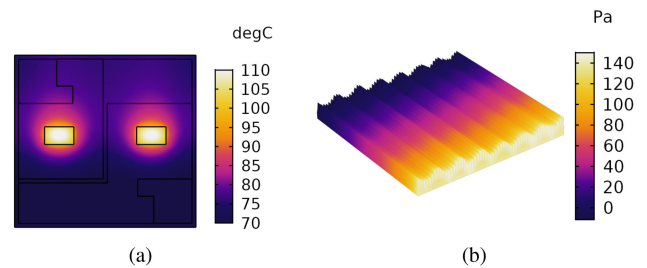


FIG. 12. (a) Steady-state thermal performance (maximum junction temperature is 110 °C) and (b) pressure drop of the selected heat sink geometry in Fig. 10.

requirement is justified, leading to a 30% increment in power density.

The pin-fin and the optimized design from the proposed approach were compared at various flow rate conditions using FEA. The objective was to compare the heat transfer performance of the two designs under different operation conditions. Constant and uniform heat load across the surface of each heat sink was applied at different flow rate conditions and the average heat transfer coefficient (htc) between the heat sink surface and coolant was calculated using

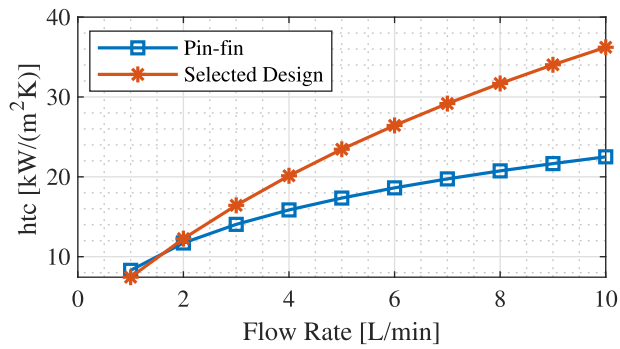
$$htc = \frac{Q}{A \cdot \Delta T} \quad (8)$$

where Q is heat transfer in Watts, A is heat sink surface area and ΔT is the temperature difference between the surface of the heat sink and the mean coolant temperature [16]. Heat transfer coefficient values for the pin-fin and selected design with respect to flow rate and pumping power at 65 °C coolant temperature are presented in Fig. 13(a) and 13(b), respectively. The selected design matched the pin-fin performance at low-flow rate conditions with negligible increase in pumping power. On the other hand, the selected design provided up to a 50% increase in thermal performance over the pin-fin design while maintaining the pumping power below 1 W as shown in Fig. 13(b).

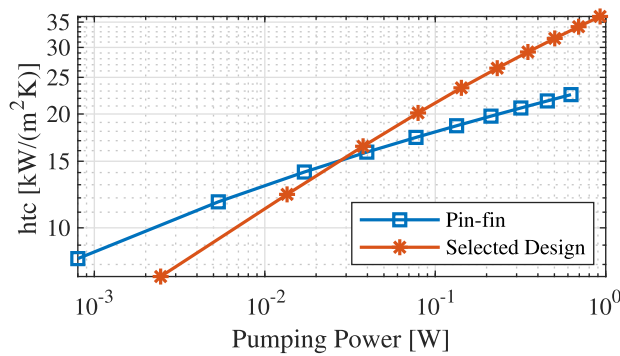
VI. EXPERIMENTAL VALIDATION

A. EXPERIMENTAL SETUP

The performance of the selected design in Fig. 10 was demonstrated in an experimental setup using a representative application scenario. Selected design in Fig. 10 was fabricated using conventional prototyping method electric discharge machining (EDM) from an aluminum block. The realized heat sink prototype is presented in Fig. 14. In proposed design approach, the designs are generated in a 2D domain and then extruded through the power module/substrate. Therefore, the developed geometry can be realized with conventional manufacturing methods and doesn't rely on additive manufacturing method, used in [7]. The design can also be realized with aluminum extrusion method, which is commonly used for high volume manufacturing. The isometric and cross-section of the experimental setup are shown in Fig. 15(a) and 15(b), respectively. The test setup consisted of a DBC structure with



(a)



(b)

FIG. 13. htc comparison of the optimized and pin-fin heat sinks with respect to (a) flow rate and (b) pumping power between 1 and 10L/min flow rate conditions.

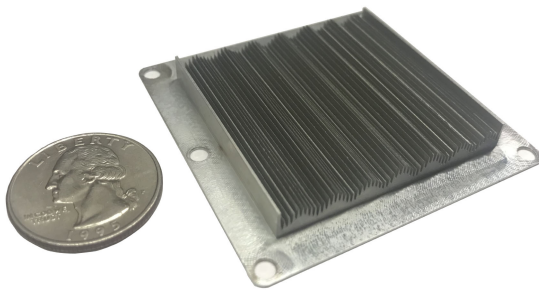
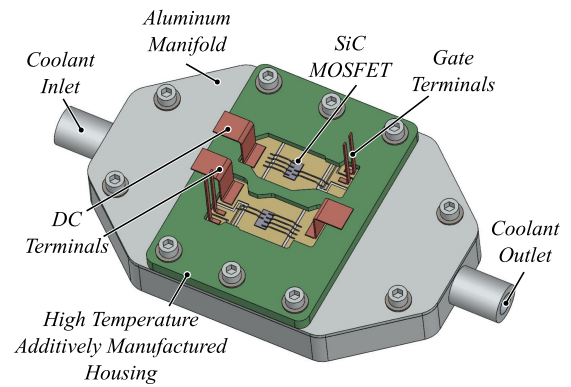
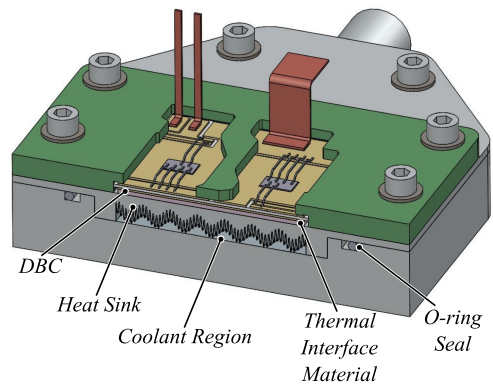


FIG. 14. Prototype of the selected design.

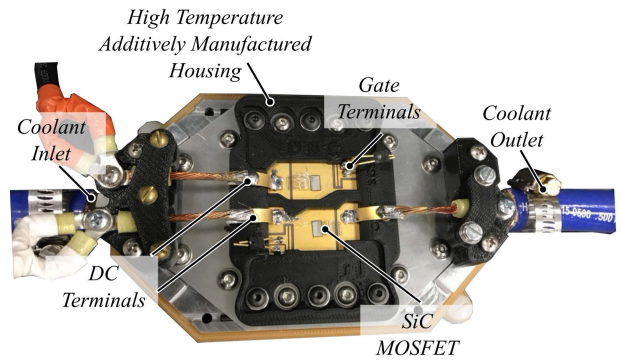
SiC MOSFETs (shown in Fig. 8), which is mounted on the optimized heat sink design shown in Fig. 11. The heat sink was machined on a manifold structure that seals the liquid cooling, seals the enclosure with O-ring seals, and evenly distributes the coolant across the heat sink coolant region.. The DBC was mounted to the heat sink using a thermal interface material and clamped using the high-temperature additively manufactured housing. As mentioned in Section V, one 900V, 10 mΩ (CPM3-0900-0010A) SiC MOSFET die per switch location is used and attached on the DBC with 50 μm thick 63-Sn/37-Pb-based solder. The DBC substrate used for the experimental validation in this study was formed by a 640 μm thick AlN ceramic insulator sandwiched between copper planes that are 300 μm thick. The thermal interface



(a)



(b)



(c)

FIG. 15. Experimental setup: (a) isometric view and (b) cross-section of the structure highlighting the heat transfer area (c) realized prototype.

material used in this study was 1 mm thick with 5 W/(mK) thermal conductivity.

During the experimental analysis, the SiC MOSFETs in Fig. 15(a) in the half-bridge configuration were turned off by applying 0 V to their gate. Then, a controllable direct current (DC) source was connected to DC- and DC+ terminals on the substrate to dissipate power loss across the devices by using relatively large on-state voltage drop of body diode of SiC MOSFETs. The power losses across the devices were measured by probing the current through the devices and the voltage across the substrate terminals via a power analyzer

TABLE I Test Conditions

Parameter	Value
Power Loss per SiC MOSFET Die	50W
Inlet Coolant Temperature	25, 45 and 65 °C
Coolant Flow Rate	1.66, 2, 4, 6L/min
Maximum Allowed Junction Temperature	130 °C

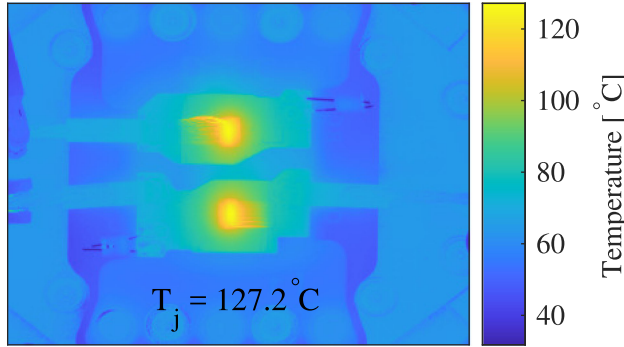


FIG. 16. Thermal image of the SiC MOSFET dies at following experimental conditions: 50 W power loss per chip, 65 °C coolant temperature, 1.66 L/min flow rate.

to ensure symmetrical power losses between top and bottom devices in the half-bridge. The total power dissipation across two SiC MOSFETs was kept at 100 W for three different coolant temperatures of 25 °C, 45 °C, and 65 °C at 1.66, 2, 4, 6L/min flow rate conditions. Test conditions are summarized in Table I. The flow rates and coolant temperatures were regulated with a closed-loop chiller. The pressure drop across the manifold was monitored via a differential pressure transducer. The substrate surface and SiC MOSFET die surface temperatures were measured using an infrared thermal camera. For this purpose, the top surface of the sample was painted with flat black spray paint to achieve constant emissivity.

B. EXPERIMENTAL RESULTS

The experimental results under different operating conditions (e.g., flow rate, coolant temperature) are presented along with finite element simulation results obtained from COMSOL. The finite element model of the experimental setup was developed using multi-physics model to validate the modeling approach used for the multi-objective optimization framework discussed in Section III. The junction to coolant thermal resistance R_{th} versus coolant flow rate based on experimental and simulation results is presented in Fig. 17. The R_{th} is defined as

$$R_{th} = \frac{T_{j-max} - T_{Coolant}}{P_{Loss}} \quad (9)$$

where T_{j-max} is maximum junction temperature of SiC MOSFET, $T_{Coolant}$ is average coolant temperature across the heat sink and P_{Loss} is the power loss across a single SiC MOSFET chip.

The temperature distribution across the substrate and SiC MOSFET dies at 50 W power loss per chip, 65 °C coolant temperature, and 1.66 L/min flow rate conditions is shown in

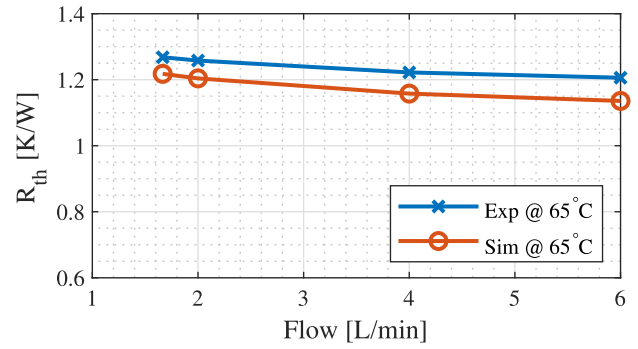
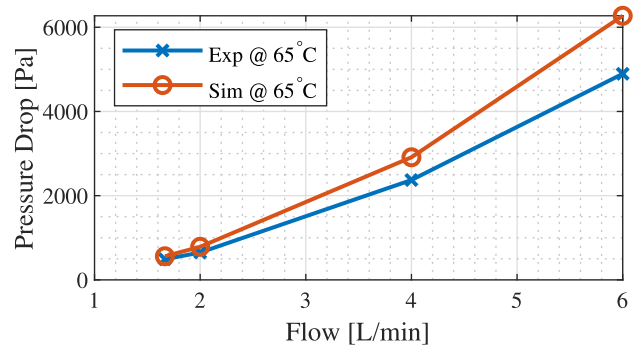
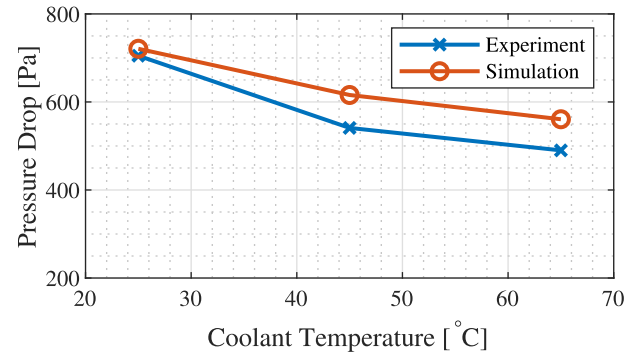


FIG. 17. Experimental and simulation results for junction to coolant thermal resistance versus flow rate at 65 °C coolant temperature. (b) Pressure drop across manifold versus flow rate at 65 °C coolant temperature.



(a)



(b)

FIG. 18. Comparison of experimental and simulation results: (a) pressure drop across manifold versus flow rate at 65 °C coolant temperature, and (b) pressure drop across manifold versus coolant temperature at 1.66 L/min flow rate.

Fig. 16. The power loss was distributed symmetrically across two dies and the R_{th} results in Fig. 17 show that the experimental and simulation results are in good agreement under various operating conditions. Although the R_{th} is dominated by the interface resistance between the bottom of the DBC and heat sink surface (due to experimental setup limitations in which direct attachment was not possible), the impact of increased heat transfer coefficient of the heat sink with flow rate predicted in Fig. 13(a) was observed in the experimental

setup, as well. Furthermore, the finite element model of the experimental setup captured the heat transfer performance within less than 5% error.

The pressure drop across the aluminum manifold shown in Fig. 15(a) was measured in the experimental setup and compared with the simulation results under different coolant temperature and flow rate conditions. The experimental and simulation results are presented in Fig. 18(a) and 18(b) with respect to a 65 °C coolant temperature at 1.66 L/min flow rate. The experimental results are well aligned with the simulation results and the trend of an increase in pressure drop with increased flow rate. On the other hand, the pressure drop reduced with an increase in coolant temperature because of a reduction in viscosity of the water-ethylene glycol mix. At low-flow rate conditions (less than 2 L/min), the difference between the experimentally observed and simulated pressure drop was less than 75 Pa under a wide range of coolant temperatures. The pressure drop measurements confirm the accuracy of the finite element models developed to generate complex heat sink geometries, and also show that even with the additional pressure drop across the manifold and cooling piping, high-performance heat sink designs with low pumping requirements are achievable.

VII. CONCLUSIONS AND FUTURE WORK

This article presents a Fourier analysis-based and GA-based constrained multi-objective design optimization of liquid-cooled heat sinks. The optimized heat sink demonstrated a significant heat sink volume reduction, which raised the power density by 30% compared with the pin-fin heat sink. The optimized heat sink also achieved better thermal performance in small form factor at high-flow rate conditions. These cooling performance improvements will help create high-power density wide-bandgap power modules. Based on the optimization results, a prototype was fabricated using conventional machining methods (EDM wiring and CNC machining) to demonstrate the performance and validate the proposed finite element-based optimization process. The experimental results show that developed finite element models can predict the thermal and flow performance of the complex design with high fidelity, and the results validate the proposed design approach. Fourier-based heat sink design procedure presented in this study is not limited to power modules, and can be used in other applications where fin based heat sinks are used for heat transfer between the source and the ambient. As part of future work, an extension of the Fourier analysis-based design to 2D and 3D structural variations will be studied.

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